Shakil Ahmed Rafi

Ph.D. Candidate,
Aspiring Data Scientist & ML Engineer.
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Passionate and hard-working graduate student working at the intersection of machine learning, deep learning, and differential equations. I aim to bring the tools of mathematics to bring value and make a difference in my community.

Education

University of Arkansas, Ph.D.
2020-Present
Pure Mathematics.
Recipient, Lawrence Jesser Toll Jr. Endowed Scholarship.
Treasurer, Graduate Student Colloquium.

University of Arkansas, M.Sc.
2016-2019
Pure Mathematics.
Vice President, Bangladesh Student Organization

Troy University, B.Sc.
2011-2015
Pure Mathematics & Philosophy.
Graduated Cum Laude.
Member, Alabama Eta chapter of Pi Mu Epsilon, the national math honor society.
Recipient of the Millennium Scholar's Award.

Employment History

PhD Candidate, University of Arkansas
2020-Present
Research area involves the use of artificial neural network in solving certain elliptic PDEs.

Graduate Teaching Assistant, University of Arkansas
2016-2020
Responsible for teaching, lesson-planning, and interfacing with a cohort of faculty in teaching 500+ students along with grading and office hours.

Asst. Business Analyst, R.S.S. Wears Ltd.
2015-2016
Implemented software for a garment manufacturing start-up, including, payroll, book-keeping with Excel, website setup, paper-trail, and staff directories.

Concessions Manager, Sodexo, Inc.
2011-2013
Set up concessions for Troy University games, with a team of six other people, including logistics, safety and customer satisfaction.

Technical Skills

Experienced in Python, including Scikit-learn, XGBoost Pandas, Folium, Geopandas, matplotlib, plotly, NumPy, Pandas, Dask, and other tools.

Proficient in SQL, Microsoft Azure, and PowerBI.

Certified Wolfram Technology Associate. Skilled in MATLAB.

Diligent, collaborative, excellent communication, and oral skills.

Project, Workshops, and Talks

SIAM Mathematics of Data Science, 2022
San Diego, CA
Who Rides Uber Anyway? Rafi & Nithila. We present a clustering approach to classifying ride shares in Chicago during 2020 and the insights it gave us. Pre-print (clickable hyperlink on PDF version on QR Code): http://doi.org/10.36277/techrxiv.21076042.v2.

Arkansas Summer Research Institute, 2022
Online
Who and where did loans go during the pandemic? Rafi. We looked at loans disbursed during the pandemic (PPP) and saw that poorer neighborhoods tended to get smaller and less frequent loans.
Funded by Arkansas NSF EPSCOR and DART.

Arkansas Spring Lecture Series, 2022
Fayetteville, AR.
Analysis of ride-sharing data for Chicago during the era of the pandemic. Rafi & Nithila. Poster session showing initial work on ridesharing in Chicago.
Funded by NSF and the University of Arkansas.

MAA Regional MathFest, 2015
Troy, AL
Turing Machines and Theorems. Expository talk at Troy University.

MAA MathFest, 2014
Portland, OR
Congruency Algorithm between two regular n-gons. Presented original research at the MAA National MathFest.

References available upon request

Certifications

IBM Data Science Specialization, *IBM*

A Coursera based course issued by IBM in collaboration with Coursera showing the fundamentals of data science as well as culminating in a real-life project predicting the efficacy of SpaceX launches.

Advanced Google Analytics, *Google*

Advanced tools and techniques of the Google Analytics platform.

Google Analytics for Beginners, *Google*

Beginner tools and techniques of the Google Analytics platform.

Wolfram Technology Associate, *Wolfram, Inc.*

Fundamentals of Mathematica and the Wolfram Cloud Platform.

Fundamentals of Digital Marketing, *The Open University & Google*

Fundamentals of marketing strategy in a digital first world, with an emphasis on Google tools

Service

Vice President of the Bangladesh Student Organization
An organization dedicated to disseminating Bangladeshi culture and helping students of Bangladeshi origin on campus to thrive and feel at home

Departmental Representative Graduate Student Council.

Representing the interests of the students of the Department of Mathematical Sciences and the Fulbright College of Arts and Sciences.

Treasurer, The Graduate Student Colloquium

Responsible for securing funding speakers when they may appear at our weekly and bi-weekly talks at the Department.

Host and Organizer, Annual Bangladesh Festival

Host and organize the annual Bangladesh Festival for five years in a row.

Talks Given

Topology and Computation, 2020

A survey about current results in the field of computational complexity results as it relates to problems in topology, including genus determination of knots, determination of incompressibility, unknot detection etc.
Graduate Student Colloquium

Heegard Splittings and Basic Theorems about Them, 2020

Expository talk about Heegard Splitting, classical results such as Waldhausen's. Heegard Genus, and modern results by Schultens, Rieck et. al.
Graduate Student Colloquium

Why Knot Theorists are Unsure about Tying Shoelaces. 2022.
Introduction to the classic result of uncertainty of additivity of crossing number of two knots, and Diao, 2003 result for torus knots
Graduate Student Colloquium

Conferences Attended

Redbud Topology Conference, 2019, 2020

Representing three states, Arkansas, Missouri, and Oklahoma dedicated to disseminating new and exciting advances in topology, particularly geometric and low-dimensional topology.

Spring Sectional Meeting of the American Mathematical Society, 2018

Representing the interests of the students of mathematics in the states of Arkansas and Oklahoma of the Southern Section of the American Mathematical Society

Symposium on Machine Learning and Dynamical Systems. The Fields Institute, Toronto, Canada, 2022

Conference hosted by the Fields Institute in Canada on the state of art in machine learning as it applies to finding numerical solutions for dynamical systems.